

AP Calc Lesson 2.1 Homework

Answer
Key

1. a. $f(-2) = -(-2)^2 + 7(-2) = -4 - 14 = -18$

b. $f(w) = -(w)^2 + 7(w) = -w^2 + 7w$

c. $f(x+h) = -(x+h)^2 + 7(x+h) = -(x^2 + 2xh + h^2) + 7x + 7h = -x^2 - 2xh - h^2 + 7x + 7h$

d. $f(x+h) - f(x) = -x^2 - 2xh - h^2 + 7x + 7h - (-x^2 + 7x) = -2xh - h^2 + 7h$

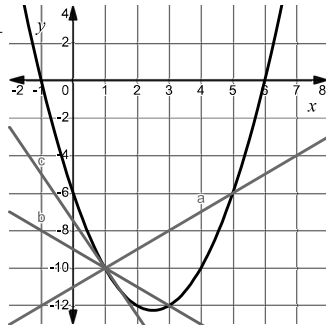
2. a. $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{(3(3+h)^2 - 8) - (3(3)^2 - 8)}{h}$
 $= \lim_{h \rightarrow 0} \frac{3(9 + 6h + h^2) - 8 - (19)}{h} = \lim_{h \rightarrow 0} \frac{27 + 18h + 3h^2 - 8 - (19)}{h}$
 $= \lim_{h \rightarrow 0} \frac{18h + 3h^2}{h} = \lim_{h \rightarrow 0} \frac{h(18 + 3h)}{h}$
 $= \lim_{h \rightarrow 0} (18 + 3h) = 18$

b. The instantaneous rate of change of f at $x = 3$ is 18. Or, the slope of the line tangent to the graph of f at $x = 3$ is 18.

3. a. Lina forgot grouping symbols in two places. She should have multiplied 4 by the quantity $(3+h)$ and the quantity $(12-9)$ should have been grouped together at the end of her work.

b. $\lim_{h \rightarrow 0} \frac{(4(3+h) - (3+h)^2) - (12-9)}{h} = \lim_{h \rightarrow 0} \frac{(12+4h) - (9+6h+h^2) - 3}{h} = \lim_{h \rightarrow 0} \frac{12+4h-9-6h-h^2-3}{h} = \lim_{h \rightarrow 0} \frac{-2h-h^2}{h} = \lim_{h \rightarrow 0} \frac{h(-2-h)}{h} = \lim_{h \rightarrow 0} (-2-h) = -2$

4. a. $\frac{f(5) - f(1)}{4} = \frac{-6 - (-10)}{4} = \frac{4}{4} = 1$



b. $\frac{f(3) - f(1)}{2} = \frac{-12 - (-10)}{2} = \frac{-2}{2} = -1$

c. $\frac{f(1.5) - f(1)}{0.5} = \frac{-11.25 - (-10)}{0.5} = \frac{-1.25}{0.5} = -2.5$

d. The value in part c, -2.5, since the two points used to calculate the slope of the secant line are very close together, and this secant line looks almost identical to the tangent line at $x = 1$. $h = 0.5$ is the closest value of h to zero.

5. G and H

6. a. $k = -3$ since this expression represents the slope of the linear function which is -3 .

b. -3 ; this expression represents the slope of the tangent line at $x = 4$ which is also -3 because the rate of change of a linear function is constant.

c. -3 ; this expression also represents the slope of the tangent line, but at $x = -2$, and the slope of the tangent line at any x -value is -3 because f is a linear function.

7. a. $f(x) = \tan x$

b. $c = \frac{\pi}{4}$

8. a. $\frac{C(3) - C(1)}{3 - 1} = \frac{\frac{20}{8} - \frac{20}{6}}{3 - 1} = \frac{-\frac{5}{6}}{2} = -\frac{5}{12} \frac{\text{mg/L}}{\text{hr}}$

b. $\lim_{x \rightarrow 2} \frac{\frac{20}{5+x} - \frac{20}{7}}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{140}{7(5+x)} - \frac{20(5+x)}{7(5+x)}}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{40 - 20x}{35 + 7x}}{x - 2} = \lim_{x \rightarrow 2} \left(\frac{-20(x-2)}{35 + 7x} \cdot \frac{1}{x-2} \right) = \lim_{x \rightarrow 2} \frac{-20}{35 + 7x} = \frac{-20}{49}$

9. D

10. B